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Rolls-Royce	ENGINEERING SPECIFICATION	RRES 90058
		Revision (Initial Issue)
		First Issued 17 Mar 2015
		Revised (New)

Cure Date Identification and Storage of Elastomers

1 SCOPE:

1.1 This specification provides the requirements for controlling the cure date identification, packaging, storage and shelf life for elastomer parts.

2 REASONS FOR USE:

2.1 To ensure that components from Elastomer materials are provided for engine build, or for delivery as customer spares, in a condition compliant with the Component Definition requirements.

3 APPLICABLE DOCUMENTS:

3.1 The following publications form a part of this document to the extent specified herein. The applicable issue shall be the current issue, unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply. Additionally, contact your sector's Rolls-Royce Technical Authority to obtain guidance/approval on action to remove or replace the cancelled reference by a drawing and/or specification change. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

3.2 Society of Automotive Engineers (SAE) Aerospace Recommended Practices (ARP). Available from SAE, 400 Commonwealth Dr., Warrendale, PA 15096, USA:

SAE ARP5316	Storage of Elastomer Seals and Seal Assemblies Which Include an Elastomer Element Prior to Hardware Assembly

3.3 ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA19428-2959, United States:

ASTM D 1418	Standard practice for Rubber and Rubber Latices - Nomenclature

4 DEFINITION OF TERMS:

Elastomer:
Fully cured (cross linked) elastic material in the form of finished parts and product.

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5 REQUIREMENTS:

5.1 The shelf life is given in Table 1 for the various classes of elastomer material in common use. Shelf life extensions are not normally recommended. These shelf lives apply only where packaging and storage has been properly carried out.

5.2 Packaging guidance for elastomers shall be as specified in ARP5316 if no other requirements for elastomers are defined on the Component Definition.

TABLE 1 - SHELF LIFE FOR CORRECTLY PACKAGED AND STORED ELASTOMERS

MATERIAL	DESIGNATION AS PER ASTM D1418*	RECOMMENDED SHELF LIFE / YEARS
Perfluororcarbon	FFKM	Unlimited
Fluorocarbon	FKM	Unlimited
Fluorosilicone	FVMQ	Unlimited
Silicone	VMQ	Unlimited
Tetrafluorethylene / propylene	FEPM	Unlimited
Perflurorether	N/A	Unlimited
Polyetherurethane	EU	5
Polyesterurethane	AU	5
Natural Rubber	NR	5
Chloroprene	CR	15
Nitrile	NBR	15
Hydrogenated Nitrile	HNBR	15
Ethylene-propylene rubber	EPDM	Unlimited
Butyl	IIR	15
Acrylate	ACM	7
Ethylene Acrylic	AEM	7
Styrenebutadiene	SBR	3

*ASTM D1418 provides a system for naming materials.

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6 NOTES:

Issue	Change (ESC) Number	Change Summary (In Descending Order)	Date
Initial	981	- EDNS04000034923	17 Mar 2015

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